Latypov Vladimir

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ML researcher and HSE × YSDA master's student with a background in mathematics, computer science, and machine learning. My current research focuses on generative modeling, especially diffusion models, GFlowNets, RL fine-tuning, and applications to protein inverse folding.

Education

2025 September – Present — **Master's student, HSE University × YSDA, Moscow,**
Applied mathematics and computer science (AI specialization).

2021 September – 2025 June — **Bachelor, ITMO University, SPb ,** *Applied mathematics and computer science (aka KT), 38th group.*
Bachelor's thesis: *Evolutionary NAS for Convolutional KANs*. Received a 5+ grade and the *Best Diploma at ITMO Award*.

2017 September – 2021 May — **Physics and Technology Lyceum named after Z.I. Alferov,** *2021A.*

Work experience

2025 December – Present — **ML Researcher,** *BayesGroup, HSE University.*

- Working on a paper on RL alignment of latent diffusion LMs.
- Made a coursework on structure-dependent learnable generation order and occupancy-aware ensemble conditioning for protein design.

2024 September – 2025 December — **ML Researcher, Maths in ML & Physics-Informed ML group of Alexander Hvatov,** *NSS Lab, ITMO University.*

- **Paper, second author**: Applying symbolic differential equations to diffusion problems for physical data. Co-led the work; my contribution focused on SINDy-based flow matching.
  - **Paper**: Evolutionary NAS for Convolutional KANs. Developed a tailored NAS algorithm and new convolutional KAN layers; initially prepared as a top-tier ML conference submission. The work served as my bachelor's thesis, won Best Paper at the ITMO Congress of Young Scientists`25, received the Best Diploma at ITMO Award, and was presented at MathAI`25.
  - Took part in the ongoing work dedicated to searching conserved quantities in differential equations with neural operator learning.

2023 December – 2024 September — **ML Researcher, AutoML and evolutionary computation group of Nikolay Nikitin,** *NSS Lab, ITMO University.*

- **Paper, Q1 Journal**: Contributed to a Knowledge-Based Systems paper on sensitivity analysis for evolutionary AutoML pipelines: concept development, experiments, and manuscript writing.
- Initiated research directions within the lab:

- Cellular encodings: novel genetic operators, integration into the GOLEM library, *theoretical results* of convergence asymptotics for model tasks based on drift theorems.
- RL-based hyperparameter adaptation for evolutionary algorithms, with handcrafted features as states and hyperparameter choices as actions.
- Contributed enhancements to the GOLEM open-source library.

2021 January – Present

Research engineer, optimal control and embedded systems, *AO NPP Kometech.*

- Lead the development of a hard real-time device for phase synchronization in communication systems using model predictive control.
- Develop hard real-time / RTOS software in Rust and C++, and FPGA modules in Clash/Haskell and Verilog; also contributed to GUI development and hardware supervision.

2020 July – 2022 February

Student Project Lead, Evolutionary Computation, *Robotics Lab, ITMO University.*

- Combining gradient-based and evolutionary optimization methods for image vectorization (my team's task, 3 students) for subsequent painting with a robotic arm (other team's task).
- Received «*ITMO.STARS*» award for this project.

Teaching

2023 September – 2024 January

Instructor for the «Rust Programming Language» Course, *Physics and Technology Liceum named after Z.I. Alferov.*

- Prepared and delivered most of the *lectures* and *practical sessions*.
- Designed homework assignments, automated tests, and reviewed student code.

The lecture recordings are available on *YouTube* and the course materials are in the *book*.

Invited to join the teaching team for an undergraduate Machine Learning course at ITMO, applied mathematics and computer science program, expected 2026.

Summer schools, projects

- ***Paper, SMILES`25***: Worked on *Categorical Iterative Proportional Fitting* at the Skoltech Summer School of Machine Learning under the supervision of Sergei Kholkin and Alexander Korotin. Published as a selected paper in *school proceedings*.
- Topological regularization based on persistent homology, course project for Topological Data Analysis at YSDA & HSE.

Professional skills

- ML/DL: generative modeling, diffusion models, GFlowNets, RL fine-tuning, AutoML, evolutionary algorithms, physics-informed ML, topological data analysis.
- Fully familiar with ***modern technologies used in ML/DL research*** including Python libraries and infrastructure tools.
- ***Research & publication pipeline***: familiar with the full scientific paper lifecycle (idea, literature review, experiments, writing, submission, rebuttal, revision), including experience at writing papers for A*/Q1.
- ***Languages***: English C2, Russian native.
- ***Embedded development***: optimal control theory, hard real-time systems, RTOS, FPGA, Rust, C++, Verilog, Clash/Haskell.
- ***Misc***: public speaking skills, typesetting systems (LaTeX, Typst), VCS, CI/CD, broad experience with programming paradigms (OOP, functional, logical, HDL, with dependent types/HoTT, async, etc.), GUI development, Linux, Vim, etc.